

Day 14 Student Handout

Geometry Summer Credit | Angle of Elevation and Depression

Name:	Date:	Period:
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Learning Targets

I can draw and label a right-triangle model from a situation. I can distinguish angles of elevation and depression. I can use trigonometry to find heights and horizontal distances and interpret the result in context.

Lesson 14A: Build the Model

Warm-Up

1.	A right triangle has angle 38 degrees and adjacent side 15. Find the opposite side.
2.	A right triangle has opposite side 12 and adjacent side 20. Find the acute angle.
3.	From a ground-level angle, which side represents the vertical height: opposite, adjacent, or hypotenuse?

Concept Launch

Sight Lines and Horizontal Lines

An angle of elevation is measured upward from a horizontal line. An angle of depression is measured downward from a horizontal line. Horizontal lines are parallel, so an angle of depression equals the alternate interior angle of elevation in the model.

Problem	Work / notes
A student stands 30 ft from a flagpole. The angle of elevation to the top is 42 degrees. Find the flagpole height.	Sketch and label: Known sides: Ratio and equation: Answer with units:
From the top of a 55-ft building, the angle of depression to a car is 28 degrees. Find the horizontal distance to the car.	Transfer the angle: Known sides: Ratio and equation: Answer with units:

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Lesson 14B | Multi-Step Trig Models

Lesson 14B: Interpret the Situation

Modeling Routine

1. Sketch the horizontal, vertical, and line of sight. 2. Mark the right angle and reference angle. 3. Label known and unknown lengths. 4. Choose a trig ratio. 5. Solve, round only at the end, and write a sentence with units.

Problem	Work / notes
A person whose eye level is 5.5 ft stands 40 ft from a tree. The angle of elevation to the top is 36 degrees. Find the total tree height.	Triangle height: Eye-level adjustment: Total height:
A rescue observer is 80 ft above the ground. The angle of depression to a marker is 32 degrees. Find the horizontal distance.	Sketch: Trig equation: Distance:
A guy wire makes a 58-degree angle with level ground and reaches the top of a 24-ft pole. Find the wire length.	Known/unknown sides: Trig equation: Wire length:

Independent Check

4.	A building casts a 35-ft shadow when the angle of elevation of the sun is 51 degrees. Find the building height.
5.	Explain why an angle of depression can be placed at ground level as an angle of elevation.